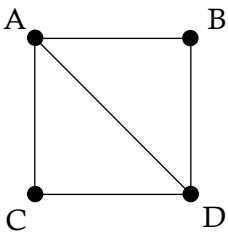


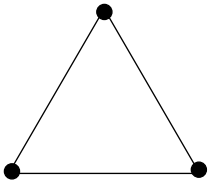
Subgraphs

A subgraph of a graph G consists of a subset of the vertices of G together with a subset of the edges of G , with the restriction that every edge in the subgraph must join two vertices in the subgraph.

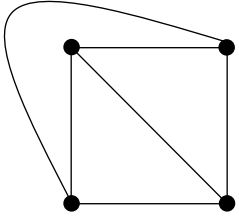
1. Draw two different, but isomorphic subgraphs of the following:

		
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2. Draw all the non-isomorphic subgraphs of K_3 . Name any graphs which are either complete, complete bipartite, or disjoint unions of such graphs. The first and last have been done for you.

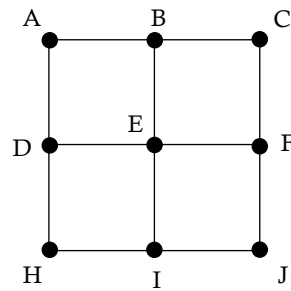
		
K_3		
	K_0 . The empty graph	

3. Draw all the non-isomorphic subgraphs of K_4 that aren't already included in your answer to Q2. Name any graphs which are either complete, complete bipartite, or disjoint unions of such graphs. One has been done for you.

		
K_4		

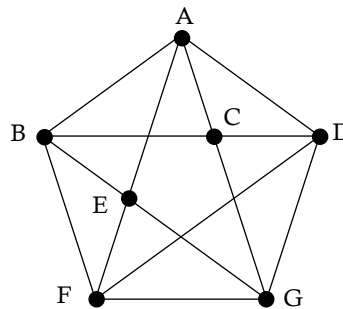
4. Identify the complement of each graph you drew in Q3. The complement of a graph G is a graph G^* which has the same set of vertices as G . Two edges are adjacent in G^* if and only if they are **not** adjacent in G . Note that $G^{**} = G$.

5. Consider the following graph:

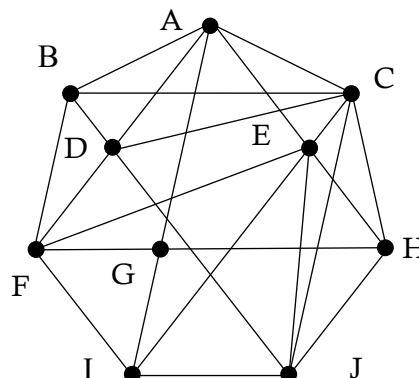


We are about to show that no subgraph is isomorphic to K_3 in three different ways:

- a. Suppose that such a subgraph exists:
 - i. A cannot be in the subgraph because it is only adjacent to B and D, and B and D are not adjacent to each other.
 - ii. Argue similarly that B and E cannot be in the subgraph.
 - iii. What property of the graph now means that none of the vertices can be in the subgraph?
 - b. Observe that the vertices of degree 2 and the vertex of degree 4 are only adjacent to vertices of degree 3. Turn this into a proof that no subgraph is isomorphic to K_3 .
 - c. Colour the vertices on the diagonals of the graph alternately red and blue. Write a proof that uses this colouring.
6. Show that no subgraph of the graph below is isomorphic to K_4 .



7.
 - a. Find a subgraph of the graph in Q6 which is isomorphic to the graph given in Q1.
 - b. How many different (correct) answers are there to part a?
8. Show that no subgraph of the following is isomorphic to K_5 .



9. Find the maximum number of edges that a graph of order 6 can have such that it does not contain a subgraph that is isomorphic to K_3 .